MANIKANTA GOPI

Founding Engineer & Senior Software Engineer — Backend, Platform & Cloud Cost
Hyderabad, India | +91 99488 95866 | gopimanikanta50@gmail.com
linkedin.com/in/gopimanikanta | github.com/ItsManikantaGopi | manikantagopi.com

SUMMARY

Founding engineer at Praja, a social app for Indian users, with 4+ years building and scaling its backend and AWS platform as the company grew — Ruby on Rails, Node.js/NestJS and Go services on AWS EKS, run through GitOps. I measure first, ship in stages, and prove impact with production metrics and the AWS bill.

HIGHLIGHTS

- Cut the AWS bill ~30% (≈$600/day → ≈$405–435/day, ~$5.5–6k/month) through Karpenter spot-first pools and caps, right-sizing and GitOps drift fixes, measured in Cost Explorer.
- Shipped on-the-fly WebP delivery for ~6.4M image requests/day: ~72% smaller images, 60–73% less CDN transfer, ~$700–760/month saved, with no backfill.
- Replaced a per-creative OpenAI o3 call with a contrast-first vision-model pipeline (~100× cheaper, 40–120 s → 5–7 s); designed a native poster renderer 6–9× faster than headless Chrome (load-tested on pre-production, in review).

TECHNICAL SKILLS

Languages: Ruby, TypeScript, Go, Python, SQL, Dart

Backend: Ruby on Rails, Node.js, NestJS, Sidekiq, BullMQ, Asynq, Socket.IO, REST APIs, JWT, Redis Pub/Sub, Redpanda

Data: Aurora MySQL, PostgreSQL, MongoDB, Redis, Dragonfly, OpenSearch, Amazon S3

Cloud & FinOps: AWS (EKS, EC2 Spot, Compute Savings Plans, CloudFront, S3, Aurora/RDS, EBS gp3, Route 53, IAM, Cost Explorer); Azure (AKS); GCP (GPU)

Kubernetes & IaC: Kubernetes, Karpenter, KEDA, HPA, Helm, Flux, ArgoCD, Terraform, Docker, GitOps, staged rollouts

Media & AI: sharp/libvips, Puppeteer/headless Chrome, Takumi, WebP/JPEG/PNG, CDN caching; OpenAI API, vision models, ONNX Runtime, PyTorch, WCAG contrast

Observability & CI: Grafana, Loki, Prometheus, Tempo, New Relic, load testing; CircleCI, GitHub Actions

PROFESSIONAL EXPERIENCE

Circleapp Online Services (Praja) — Hyderabad, India | Founding Engineer | November 2021 – Present

Joined as one of Praja's founding engineers and grew with the company from intern to Senior Software Engineer, building and scaling its backend, real-time, media and infrastructure platform.

Senior Software Engineer | April 2025 – Present

AWS / EKS cost optimization | Oct–Nov 2026 · live

- Cut the AWS bill ~30%, from ≈$600/day ($18.6k in October) to ≈$405–435/day (~$5.5–6k/month); EC2 compute alone fell from ≈$120/day to $35–60/day.
- Capped Karpenter's public node pool at 64 vCPU and moved stateless services to spot-first pools after an autoscaling burst added 192 vCPU of on-demand capacity (~$740 extra in 5 days); right-sized Redpanda, OpenSearch (6 → 4 m6g.4xlarge), Tempo retention and Jenkins, and removed all paid gp3 IOPS/throughput (−$265/month).
- Found Helm/Flux drift (a leftover nodeSelector pinning the main API to on-demand), audited all 60 Helm releases, and validated every change against Cost Explorer, the private CloudFront rate card and Savings Plan coverage.

WebP image delivery pipeline | Oct–Nov 2026 · live

- Added on-the-fly WebP conversion to the Node.js image service on EKS behind CloudFront, cutting average image size ~72% (≈200 KB → ≈55 KB) and CDN transfer 60–73% across ~6.4M requests and ~1.45 TB/day; saves ~$700–760/month at our negotiated rate, with no backfill or re-upload.
- Built a shared Rails WebP URL layer for photo, video-thumbnail and admin-media models (originals kept for pixel-processing jobs), rolled out internal → 10% (user-ID bucketing + cohort analytics) → 100%, scaled for the cold cache, and fixed a launch regression within hours.

AI font & colour ranking for posters (OpenAI o3 → JEV) | Oct 2026 · live default

- Replaced a per-creative OpenAI o3 call with a "measure first, then ask the model" Rails/Sidekiq pipeline: exact WCAG 2.1 contrast on real pixels, a diverse 24-candidate shortlist, one batched vision-model request; ~100× cheaper (~$0.074 → ~$0.0007 per creative, ~$383 → ~$4/month) and 40–120 s → 5–7 s.
- Raised WCAG-passing picks from 4.2 to 4.5 of 5 and colour variety from 2.6 to 3.6 families (30 creatives, side by side); fixed a Sidekiq OOM (~450 MB per ranking) and DB pool exhaustion, and split CPU- and network-bound workers via Redis.

Realtime poster rendering (RTM) | Nov 2026 · PRs in review, load-tested on pre-production

- Designed and built on-demand rendering to replace a batch pipeline (~1.93M posters in 2 days, ~72 avg / 375 peak pods, ~28.5 s per poster, every 576 KB PNG kept in S3): JWT-signed per-user URLs from Rails, rendered on first request and cached at the CDN.
- Built a NestJS renderer with a browser-free path (Takumi + sharp): ~9–15 ms vs ~114–300 ms per poster with Puppeteer, 6–9× throughput at ~1/6 memory; Indic-script fonts, 503 + Retry-After backpressure, 54 specs. Pre-prod: 75,000 posters at ~57–71/s, 100% success; projected ~$91 → ~$5 per 1M posters.
- Load test (398k eligible users in one state): ~23 posters/s per 2-CPU media pod, 12–19 ms URL generation; projected savings of ~$2.1k/month in compute plus ~$400/month of S3 growth avoided once rolled out.

Platform ownership | 2025 – present

- Own the GitOps repository for ~24 production services on AWS EKS; moved video processing from Lambda + FFmpeg to KEDA-scaled workers (~$18,000 → ~$1,800/month); migrated Redis to Dragonfly with zero downtime; replaced a paid vision API with a self-hosted ONNX GPU service validated on ~200 production images.
- Earlier cost work: ~$500/month off database spend by fixing resource-intensive queries, ~$800 saved in the first month with S3 expiry for ephemeral media, and ~$400/month from the first Karpenter Spot + On-Demand setup.
- Lead production troubleshooting across Rails, MySQL, Redis/Dragonfly, workers, Kubernetes and networking using Grafana, Loki, Prometheus, Tempo and New Relic.

Software Engineer | May 2022 – March 2025

- Led the production migration from Azure AKS to AWS EKS with no planned maintenance window: Terraform, service-by-service cutover, Route 53 and CloudFront moves, then decommissioning.
- Built and owned the real-time messaging service (NestJS, Socket.IO, Redis Pub/Sub, MongoDB, BullMQ), moving WebSocket traffic off the main Rails app across horizontally scaled instances.
- Delivered features across a Rails app of ~185 models, 211 background workers and 28 Sidekiq queues; implemented subscription and payment lifecycles with idempotent handling of duplicate gateway callbacks.
- Built the asynchronous poster and video generation pipeline for paid features (NestJS, FFmpeg, Puppeteer, Redis, S3) and a Go notification service (Gin, GORM, Asynq, FCM) for targeted push by district, state, party and circle.

Software Engineer Intern (founding team) | November 2021 – April 2022

- Built Flutter messaging features (conversation types, message deletion, link and post previews, member lists) and the Rails API endpoints behind them.

Continual Engine — Machine Learning Intern | 2021

- Built image-to-text models in PyTorch for scientific and chemical content, using autoencoders for feature extraction.

SELECTED PROJECTS

sidekiq-assured-jobs — Open-source Ruby gem that re-enqueues in-flight Sidekiq jobs lost when a worker is killed.
github.com/praja/sidekiq-assured-jobs

rails2x — AST-based Rails → Go migration engine (tree-sitter, Goravel) generating models, controllers, workers and policies.
github.com/ItsManikantaGopi/rails2x

Seeker — Search engine from first principles: inverted index, BM25, Levenshtein automata, FSTs, BKD trees; ~20,000 lines of TypeScript with a 43-chapter book.

EDUCATION

B.Tech, Computer Science — Rajiv Gandhi University of Knowledge Technologies, Nuzvid | 2022 | GPA: 9.3 / 10